

Visualisation in Data Science

2025/2026

Project instructions

Air quality in Madrid

Problem statement

In recent years, high levels of pollution during certain dry spells in Madrid have forced the authorities to take measures to curb it, such as limiting the use of cars in the city centre. You have been hired by Madrid's city council to analyse the data and inform the population about pollution in the city. Here are some of the questions councillors have asked:

- How has pollution in Madrid evolved between 2001 and 2018?
- Which are the areas of Madrid where pollution is highest / lowest in 2018?
- Which are the areas of Madrid where pollution has improved / worsened more between 2008 and 2018?
- How have the different measurements of pollution evolved between 2008 and 2018?

Use these questions as a guide but feel free to come up with questions of your own that you feel are important to answer with the data.

Data

The available data consists of one file with station information (id, name, address, longitude, latitude and elevation) and 18 files containing data from measurements of each year between 2001 and 2018. Each measurement file has a column with the timestamp, one with the station id, and a collection of columns with the actual measurements of different types of particles and gases indicative of pollution. Please note that different files contain different measurement types.

Here's a list of measurements taken in the stations and their description.

- **S0_2**: sulphur dioxide level measured in $\mu\text{g}/\text{m}^3$. High levels of sulphur dioxide can produce irritation in the skin and membranes, and worsen asthma or heart diseases in sensitive groups.
- **C0**: carbon monoxide level measured in mg/m^3 . Carbon monoxide poisoning involves headaches, dizziness and confusion in short exposures and can result in loss of consciousness, arrhythmias, seizures or even death in the long term.
- **N0**: nitric oxide level measured in $\mu\text{g}/\text{m}^3$. This is a highly corrosive gas generated among others by motor vehicles and fuel burning processes.

- **NO₂**: nitrogen dioxide level measured in $\mu\text{g}/\text{m}^3$. Long-term exposure is a cause of chronic lung diseases, and are harmful for the vegetation.
- **PM_{2.5}**: particles smaller than $2.5\ \mu\text{m}$ level measured in $\mu\text{g}/\text{m}^3$. The size of these particles allow them to penetrate into the gas exchange regions of the lungs (alveolus) and even enter the arteries. Long-term exposure is proven to be related to low birth weight and high blood pressure in newborn babies.
- **PM₁₀**: particles smaller than $10\ \mu\text{m}$. Even though they cannot penetrate the alveolus, they can still penetrate through the lungs and affect other organs. Long term exposure can result in lung cancer and cardiovascular complications.
- **NO_x**: nitrous oxides level measured in $\mu\text{g}/\text{m}^3$. Affect the human respiratory system worsening asthma or other diseases, and are responsible for the yellowish-brown color of photochemical smog.
- **O₃**: ozone level measured in $\mu\text{g}/\text{m}^3$. High levels can produce asthma, bronchitis or other chronic pulmonary diseases in sensitive groups or outdoor workers.
- **TOL**: toluene (methylbenzene) level measured in $\mu\text{g}/\text{m}^3$. Long-term exposure to this substance (present in tobacco smoke as well) can result in kidney complications or permanent brain damage.
- **BEN**: benzene level measured in $\mu\text{g}/\text{m}^3$. Benzene is a eye and skin irritant, and long exposures may result in several types of cancer, leukaemia and anaemias. Benzene is considered a group 1 carcinogenic to humans by the IARC.
- **EBE**: ethylbenzene level measured in $\mu\text{g}/\text{m}^3$. Long term exposure can cause hearing or kidney problems and the IARC has concluded that long-term exposure can produce cancer.
- **MX_Y**: *m*-xylene level measured in $\mu\text{g}/\text{m}^3$. Xylenes can affect not only air but also water and soil, and a long exposure to high levels of xylenes can result in diseases affecting the liver, kidney and nervous system (especially memory and affected stimulus reaction).
- **PX_Y**: *p*-xylene level measured in $\mu\text{g}/\text{m}^3$. See **MX_Y** for xylene exposure effects on health.
- **OX_Y**: *o*-xylene level measured in $\mu\text{g}/\text{m}^3$. See **MX_Y** for xylene exposure effects on health.
- **TCH**: total hydrocarbons level measured in mg/m^3 . This group of substances can be responsible for different blood, immune system, liver, spleen, kidneys or lung diseases.
- **CH₄**: methane level measured in mg/m^3 . This gas is an asphyxiant, which displaces the oxygen animals need to breathe. Displaced oxygen can result in dizziness, weakness, nausea and loss of coordination.
- **NMHC**: non-methane hydrocarbons (volatile organic compounds) level measured in mg/m^3 . Long exposure to some of these substances can result in damage to the liver, kidney, and central nervous system. Some of them are suspected to cause cancer in humans.

More information on the data can be found at its source:

<https://www.kaggle.com/datasets/decide-soluciones/air-quality-madrid>

Questions

Please ask questions about the project on the Discussion boards in Blackboard. Other questions can be directed to me via email: inigo.bermejo@uhasselt.be.